

CAPSTONE PROJECT

Business Data Management

SOHAM BHATTACHARYA
21f3001729

Mid-Term Submission
21f3001729@ds.study.iitm.ac.in

Title: Analyzing the Revenue Trend & Sale Volume and exploring possibilities of boosting the existing revenue generation of a stationery shop

Executive Summary

As mentioned in the proposal, a 25-year old local stationery shop named 'Nandi Gift House' was approached for the purpose of this project. It primarily sells stationery items along with cosmetic goods and some food items. Mr. Shantanu Nandi is the owner of the shop. Currently, it is being managed and taken care of by his sons; primarily his eldest son, Mr. Abhishek Nandi. It has an appreciable reputation in the locality for the flawless services they have been providing so long. Mr. Nandi provided me with the raw data on 26th December, 2022. He usually maintains the register for the business dealings and sales. Data for a period of 4 months (August to November) were provided. Recently, they started to maintain their daily data in digital format i.e. in Excel Spreadsheet. So, the data received was in an Excel file. A short interaction video was also shot with Mr. Abhishek.

Initially, data cleaning was required since the received raw data was not properly formatted and was found to contain noise. Using the available MS Excel tools, data parsing was done as well.

Next, according to the objective, the data was examined to analyze the different scopes to conclude solutions for the problems using different analysis techniques and tools from Excel.

Proof of originality of the data

Link of the raw data and its corresponding analysis : [Nandy Gift House - 26 December 2022.xlsx](#)

Interaction video link :
https://drive.google.com/file/d/1ClfPtmtLMS5AGrwsKPv6F70IxcEYTbSA/view?usp=share_link

Problem Statement

- Maximizing the existing revenue generation. On a glimpse of the data received for one month, it is found that for a few categorical items, there is a possibility for increasing the revenue.
- Since, there exists a stabilized reputation of the shop for its services, we will explore the possibilities of introducing a new product which will increase the profit of the business.

The organization seems to hold a valuable reputation to its customers. Besides, it is found after visiting the shop for the interview, there are very few shops located nearby which sells such a variety of products as are sold by ‘Nandi Gift House’. It is being aimed to thoroughly analyze the available data of the shop and present a suitable solution to expand the business in terms of revenue by some additions and alterations of the existing inventory items.

Metadata and Descriptive Statistics

All the metadata and descriptive statistics are given referring to the raw data whose link has been attached in this document.

The sheet named “Daily Sales (Raw data)” is the modified cleaned spreadsheet which was received from the owner of the shop, Mr. Nandi. The sheet shows the data for a period of four months (August 2022 to November 2022). It contains the daily sales data. The columns store the sales record and the rows contain the dates of the sales. Column heads

show the name of the product being sold with its respective category to which it belongs. There are 29 products belonging to 9 different categories. This sheet is the primary source of data on which the entire analysis is based.

The sheet named “Rate Chart” displays the price of each product along with its categories.

The sheet named “Categorical Daily Sales” displays the revenue generated by each product of each category on a daily basis.

The next sheet named “Pivot Table” is the sheet containing the pivot table created using the source data in “Daily Sales (Raw Data)” to easily get the summary data for different results and findings. This is done using MS Excel tools.

Detailed explanation of Analysis Process/Method

The Analysis of the source data has been done in several worksheets of the workbook. All analysis has been done using the available tools in MS Excel.

Using the pivot table, different kinds of tables have been created for the purpose of analyzing the data.

Referring to the pivot table, the categorical revenue table has been extracted. The rows have been set to “Category” and the values have been set to “Revenue”. With the help of this extracted table, a bar chart has been created to graphically view the levels of revenue for each category. The categorical revenue chart has been created by selecting the categorical revenue table followed by selecting the option “Column Chart” from the tool named “chart” available in the “insert” tab.

Categories	Revenue
Beverages	₹681,280.00
Biscuits	₹179,316.00
Cosmetics	₹582,815.00
Snacks	₹1,243,449.00
Writing Media (Eraser)	₹57,456.00
Writing Media (File)	₹213,324.00
Writing Media (notebooks)	₹426,575.00
Writing Media (Pen)	₹91,080.00
Writing Media (Pencil)	₹496,932.00
Total	₹3,972,227.00

Referring to the previous table, the top 5 revenue categories were filtered out and extracted in a separate table.

Top 5 revenue generating categories	
Snacks	₹1,243,449.00
Beverages	₹681,280.00
Cosmetics	₹582,815.00
Writing Media (Pencil)	₹496,932.00
Writing Media (notebooks)	₹426,575.00

Based on the already created pivot table, the monthly sales for each category along with the daily average sales were calculated. The rows have been set to “Category” and the values have been set to “Sales” and the columns were set to “Months”. The total sales have been calculated using the figures available in the entries of this extracted table. Using the column of “Total Sales”, the daily average sales of each category has been calculated. The formula for the daily average sales is total sales for the category divided by total number of days available in the data. Using the column of “Daily Average Sales”, a pie chart has been created to get an overview of the percentage of contribution in the daily sales for each category. The pie chart has been created by selecting the column of “Daily Average Sales” followed by selecting the option “Pie Chart” from the tool named “Chart” available in the “insert” tab.

Category	August	November	October	September	Total Sales	Daily Average Sales
Beverages	4444	4291	4640	4370	17745	145
Biscuits	1362	1386	1292	1338	5378	44
Cosmetics	1805	1902	1986	1700	7393	61
Snacks	7150	6547	6766	6605	27068	222
Writing Media (Eraser)	373	397	437	389	1596	13
Writing Media (File)	2586	2416	2510	2487	9999	82
Writing Media (notebooks)	3408	3267	3376	3325	13376	110
Writing Media (Pen)	2389	2342	2315	2062	9108	75
Writing Media (Pencil)	4465	4106	4363	4084	17018	139
Grand Total	27982	26654	27685	26360	108681	

Also, selecting the “Total Sales” column, a column chart is created by selecting the tool named “Chart” from the “insert” tab in the same way as that of categorical revenue chart. The chart will facilitate the analysis as it gives a graphical summary of the total sales for each category.

From the categorical monthly sales table, the top five selling products have been extracted from the categorical monthly sales chart.

Next, using the pivot table, the amount of revenue generated day wise is extracted. From the pivot table, the rows are set to “Day” (days of the week) and the values are set to “Revenue” summarized as sum of revenues. Now from this available table, the average revenue generated per day is calculated.

Day	SUM of Revenue	Average Revenue
Mon	593048	4861
Tue	578951	4746
Wed	576214	4723
Thu	551994	4525
Fri	535520	4390
Sat	578115	4739
Sun	558385	4577
Grand Total	3972227	

Besides having the table of values, a bar chart is also created using the tools of MS Excel as mentioned earlier. The bar chart will help us clearly analyze the trend of revenue generation over the entire period based on the days of the week. To get a daily average revenue generation based on

the days of the week, a separate table has also been extracted using the pivot table in a similar fashion.

Next, using the same pivot table, the categorical monthly sales and revenue table has been extracted. In both the cases, the rows have been set to “Category”, columns set to “Months”. The only difference lies in the values which are set to “Sales” in case of the sales data, and “Revenue” in case of revenue data.

Using these two tables, a line chart has been created to analyze the flow of revenue generation and sales output for each category over the period of these four months that is from August to November. It basically displays the trend-line that is followed by the category over the entire period. These two graphical charts are named as “Categorical Sales Trend” and “Categorical Revenue Trend”.

Categorical Revenue Contribution in each month					
Category	August	November	October	September	Average
<i>Beverages</i>	16.59%	17.19%	17.52%	17.31%	17.15%
<i>Biscuits</i>	4.47%	4.82%	4.22%	4.57%	4.52%
<i>Cosmetics</i>	13.57%	15.76%	15.71%	13.66%	14.68%
<i>Snacks</i>	32.76%	29.52%	30.57%	32.31%	31.29%
<i>Writing Media (Eraser)</i>	1.32%	1.49%	1.54%	1.43%	1.45%
<i>Writing Media (File)</i>	5.47%	5.35%	5.26%	5.41%	5.37%
<i>Writing Media (notebooks)</i>	10.68%	10.91%	10.46%	10.92%	10.74%
<i>Writing Media (Pen)</i>	2.35%	2.44%	2.26%	2.11%	2.29%
<i>Writing Media (Pencil)</i>	12.78%	12.52%	12.46%	12.28%	12.51%
Grand Total	100.00%	100.00%	100.00%	100.00%	100.00%

Monthly Revenue Contribution for each category				
Category	August	November	October	September
<i>Beverages</i>	24.73%	24.19%	26.29%	24.80%
<i>Biscuits</i>	25.34%	25.74%	24.05%	24.87%
<i>Cosmetics</i>	23.64%	25.91%	27.57%	22.88%
<i>Snacks</i>	26.75%	22.75%	25.14%	25.36%
<i>Writing Media (Eraser)</i>	23.37%	24.87%	27.38%	24.37%
<i>Writing Media (File)</i>	26.02%	24.02%	25.19%	24.77%
<i>Writing Media (notebooks)</i>	25.43%	24.51%	25.07%	24.99%
<i>Writing Media (Pen)</i>	26.23%	25.71%	25.42%	22.64%
<i>Writing Media (Pencil)</i>	26.11%	24.14%	25.64%	24.12%

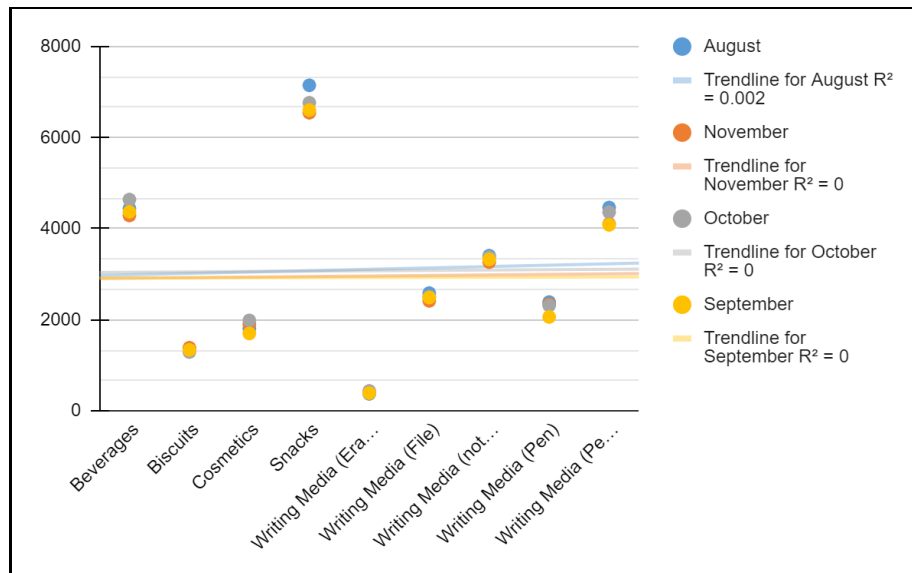
The above two tables have been calculated in order to compare and contrast and understand the impact of revenue based on the time period i.e. month of sale and revenue. With the help of these it will be easier to figure out the alterations required for the development of the business. Similar to these tables, the following tables will help to understand the impact of sales.

Categorical Sales Contribution in each month					
Category	August	November	October	September	Average
<i>Beverages</i>	15.88%	16.10%	16.76%	16.58%	16.33%
<i>Biscuits</i>	4.87%	5.20%	4.67%	5.08%	4.95%
<i>Cosmetics</i>	6.45%	7.14%	7.17%	6.45%	6.80%
<i>Snacks</i>	25.55%	24.56%	24.44%	25.06%	24.90%
<i>Writing Media (Eraser)</i>	1.33%	1.49%	1.58%	1.48%	1.47%
<i>Writing Media (File)</i>	9.24%	9.06%	9.07%	9.43%	9.20%
<i>Writing Media (notebooks)</i>	12.18%	12.26%	12.19%	12.61%	12.31%
<i>Writing Media (Pen)</i>	8.54%	8.79%	8.36%	7.82%	8.38%
<i>Writing Media (Pencil)</i>	15.96%	15.40%	15.76%	15.49%	15.65%
Grand Total	100.00%	100.00%	100.00%	100.00%	100.00%

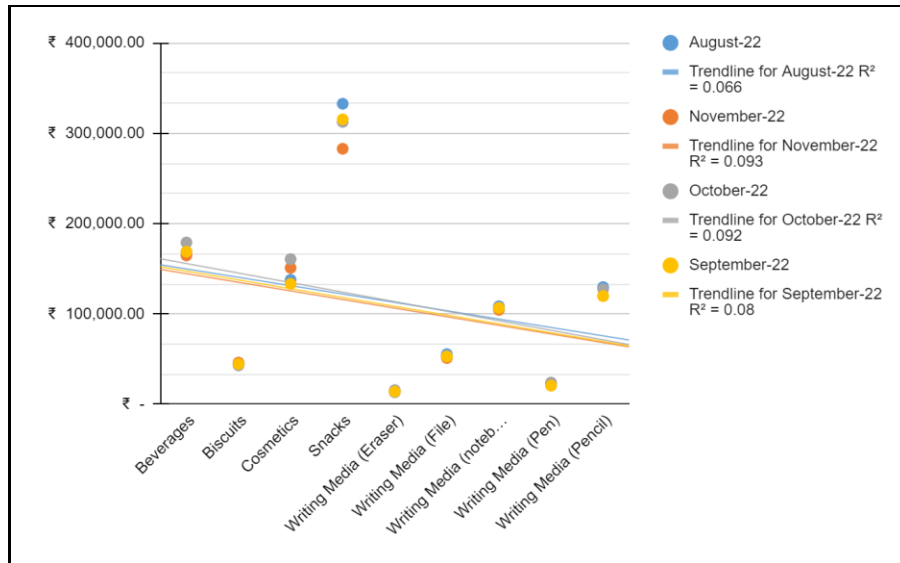
Monthly Sales Contribution for each category					
Category	August	November	October	September	
<i>Beverages</i>	25.04%	24.18%	26.15%	24.63%	100.00%
<i>Biscuits</i>	25.33%	25.77%	24.02%	24.88%	100.00%
<i>Cosmetics</i>	24.41%	25.73%	26.86%	22.99%	100.00%
<i>Snacks</i>	26.41%	24.19%	25.00%	24.40%	100.00%
<i>Writing Media (Eraser)</i>	23.37%	24.87%	27.38%	24.37%	100.00%
<i>Writing Media (File)</i>	25.86%	24.16%	25.10%	24.87%	100.00%
<i>Writing Media (notebooks)</i>	25.48%	24.42%	25.24%	24.86%	100.00%
<i>Writing Media (Pen)</i>	26.23%	25.71%	25.42%	22.64%	100.00%
<i>Writing Media (Pencil)</i>	26.24%	24.13%	25.64%	24.00%	100.00%

Two Scatter Charts have been plotted in “Sales Analysis” and “Revenue Analysis” worksheets of the attached Primary Data Excel Workbook, which will help us understand the relationship between the sales figures and the revenue figures.

Results and Findings



From the above scatter plot of the sales distribution across categories, we find that biscuits, cosmetics, erasers and snacks are considerably farther from the trend-line. As a result, these are the outliers; more specifically, categories namely biscuits, cosmetics and writing media (erasers) are low-selling products in the time period of these 4 months as mentioned in the graph; whereas, snacks occupy the maximum portion of the total sales in these 4 months.



From the above scatter plot of revenue distribution across categories, it is noticed that categories namely biscuits, snacks, writing media (erasers), writing media (file), writing media (pen). Among these outliers, snacks produce the major percentage of the total revenue generation in this time period; whereas the others provide least contribution in the revenue figures.

Now if both the graphs are simultaneously looked into, categorically we find that:

1. **Beverages** are outliers in both the graphs and are much farther away from the monthly trend-lines which imply that revenue generated by beverages do not vary or are almost stagnant per month.
2. **Biscuits** perform similar to beverages. The revenue contribution in this time period do not vary significantly, rather revenue contribution is the least in the net revenue earned in this period of 4 months.
3. We find **cosmetics** to be an outlier in the scatter plot graph of sales but not of revenue. Cosmetics, contribute significantly less in the total sales figures, yields a considerable amount of revenue.

Increasing the stock of cosmetics or introducing more products of cosmetics will result in an increase of revenues.

4. **Snacks** are one of the outliers in both the scatter plots. Consequently, they are found to yield stagnant yet high revenues in this time period.

5. Writing Media

- a. **Pencils** are found to be closer to the trend-lines in both the plot which evidently implies that there's a somewhat constant generation of revenue figures in this time period.
- b. **Erasers** are found to have the minimum contribution in the revenue generation. It is also known that pencils and erasers are complementary goods, so an increase of sales for one of them will result in the increase of sales of the other. Increasing the stock of erasers will definitely avoid a shrink in net revenue.
- c. It is found that though in sales figures, **pens** are quite close to the monthly trend-lines, their revenue contribution is considerably very less since in the revenue scatter plot, they are found to be outliers.
- d. **Files** are always a useful tool for anybody. Similar to the case of pens, it is found that the revenue contribution is quite less with respect to the sale figures. An increase in the stock of files is suggested.
- e. **Notebooks** are found to be provide a stable performance in the net revenue generation.